Chapter 1 – Introduction

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February 2017

1 Introduction

1. Welcome to MATH 5045 Advanced Analytic Techniques 1.
2. In this course we will examine data that is collected serially, for example, data collected daily, or monthly or yearly etc.
3. Such data are called Time Series.
4. In this course, we will examine the different ways of categorising Time Series. For example, is there a trend? Are there any seasonality effects?
5. We will also look at different Time Series models. Which model is most suitable for the data?
6. Finally, after fitting the appropriate model to the time series data, can we use the model to predict where the time series is heading in the near future?
7. There are widespread applications for time series modelling. Many data such as rainfall data, daily temperatures, stock prices, can be modelled using some of the standard time series models.

2 Using R

1. In this course, we will be using **R** .
2. If you do not yet have **R** installed in your computer, please go to the website www.r-project.org to download a suitable version for your personal computer. You only need the basic package. You can add the additional add-ons later if these are going to be needed for your other modules.
3. When you first open up **R** , you will see the **R** Console with a prompt >.
4. Type all your commands at the prompt and press "Enter".
5. The **R** package is already loaded with sample data.

   ```
   > data()
   ```

   This will list all the data (with description) that is already preloaded with the **R** package. Not all of these are time series data. But quite a lot of them are.
6. We will be making use of some of these plus time series data from elsewhere.
7. At any time, if you need help with the optional parameters of specific commands, type "help( )" with the command in the parenthesis (). For example

   ```
   > help(plot)
   ```

   will result in a window in your internet browser popping up displaying all the information about the plot() command. Note that you need to be connected to the internet for this to work.

3 Examples of Time Series Data

1. We now look at some of the preloaded data in **R** that are time series.

2. **Example 1**. We note that there is the airmiles data which is described as "Passenger Miles on US Commercial Airlines, 1937 – 1960". Since the data is called airmiles, we can simply type "airmiles" at the prompt

   ```
   > airmiles
   Time Series:
   Start = 1937
   End = 1960
   Frequency = 1
    [1]   412   480   683  1052  1385  1418  1634  2178  3362  5948  6109  5981
   [13]  6753  8003 10566 12528 14760 16769 19819 22362 25340 25343 29269 30514
   ```

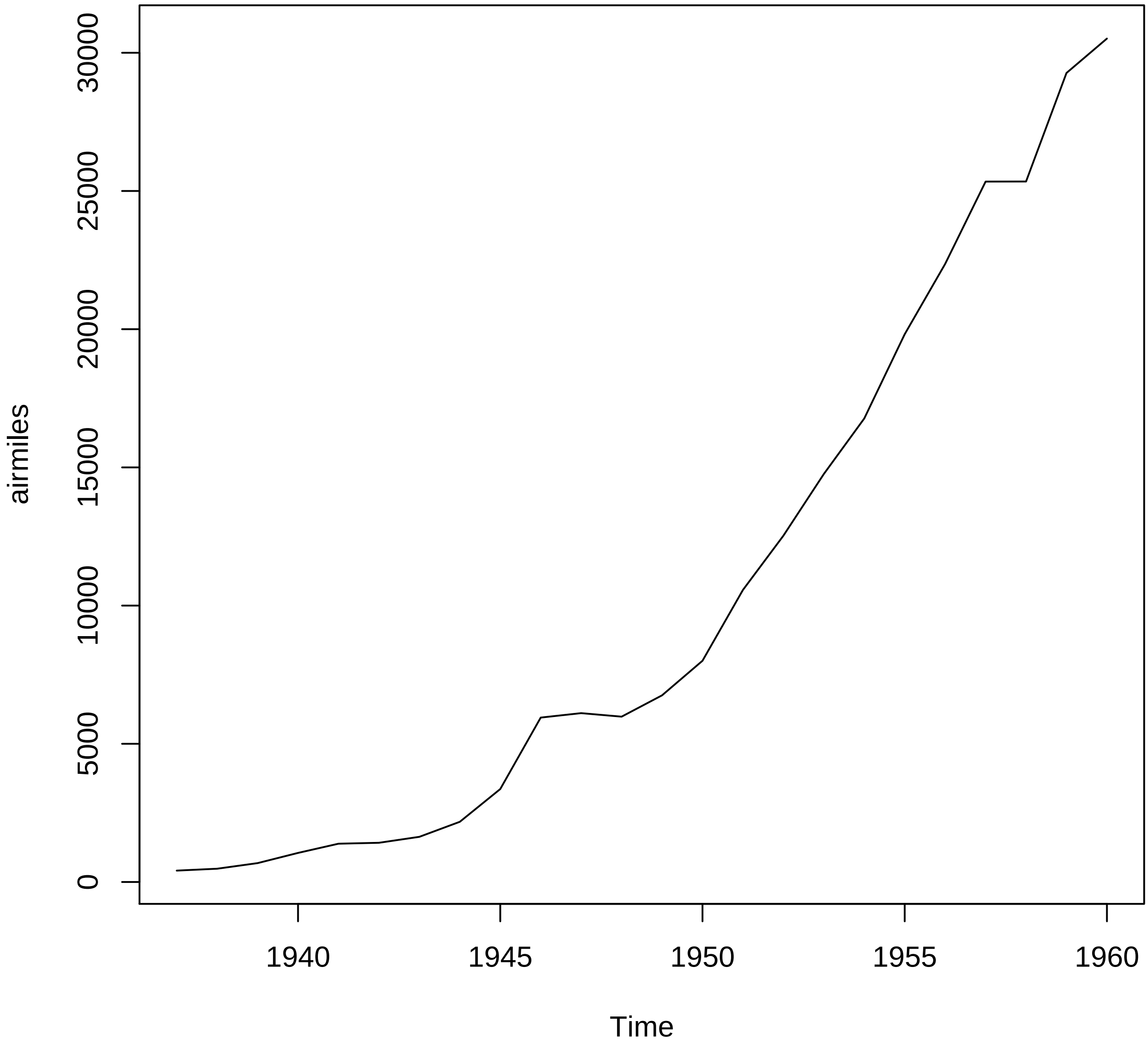
   Note that the [1] and [13] are not part of the actual data. The way **R** displays data is that [1] denotes that the 1st observation is the 1st on the 1st row and it is 412. The [13] denotes that the 13th observation is 6753.

   We can also plot the data. Since the data is already set up in **R** as a time series, we can use the plot command without any of the additional options. Type

   ```
   > plot(airmiles)
   ```

   and we get the following graph in a separate window. If you want to save the plot, activate the plot window and look under the File menu at the top where you can save it in a variety of formats such as .pdf or .eps files. From Figure 1, we can see that passenger airmiles increase over time, that is, more and more Americans are flying probably due to more planes and cheaper airfares over time.

Figure 1: Plot of US Passenger Airmiles Flown



3. **Example 2**. Average monthly temperatures at Nottingham, 1920-1939. This is stored in the dataset nottem.

```
> nottem
      Jan  Feb  Mar  Apr  May  Jun  Jul  Aug  Sep  Oct  Nov  Dec
1920 40.6 40.8 44.4 46.7 54.1 58.5 57.7 56.4 54.3 50.5 42.9 39.8
1921 44.2 39.8 45.1 47.0 54.1 58.7 66.3 59.9 57.0 54.2 39.7 42.8
1922 37.5 38.7 39.5 42.1 55.7 57.8 56.8 54.3 54.3 47.1 41.8 41.7
1923 41.8 40.1 42.9 45.8 49.2 52.7 64.2 59.6 54.4 49.2 36.3 37.6
1924 39.3 37.5 38.3 45.5 53.2 57.7 60.8 58.2 56.4 49.8 44.4 43.6
1925 40.0 40.5 40.8 45.1 53.8 59.4 63.5 61.0 53.0 50.0 38.1 36.3
1926 39.2 43.4 43.4 48.9 50.6 56.8 62.5 62.0 57.5 46.7 41.6 39.8
1927 39.4 38.5 45.3 47.1 51.7 55.0 60.4 60.5 54.7 50.3 42.3 35.2
1928 40.8 41.1 42.8 47.3 50.9 56.4 62.2 60.5 55.4 50.2 43.0 37.3
1929 34.8 31.3 41.0 43.9 53.1 56.9 62.5 60.3 59.8 49.2 42.9 41.9
1930 41.6 37.1 41.2 46.9 51.2 60.4 60.1 61.6 57.0 50.9 43.0 38.8
1931 37.1 38.4 38.4 46.5 53.5 58.4 60.6 58.2 53.8 46.6 45.5 40.6
1932 42.4 38.4 40.3 44.6 50.9 57.0 62.1 63.5 56.3 47.3 43.6 41.8
1933 36.2 39.3 44.5 48.7 54.2 60.8 65.5 64.9 60.1 50.2 42.1 35.8
1934 39.4 38.2 40.4 46.9 53.4 59.6 66.5 60.4 59.2 51.2 42.8 45.8
1935 40.0 42.6 43.5 47.1 50.0 60.5 64.6 64.0 56.8 48.6 44.2 36.4
1936 37.3 35.0 44.0 43.9 52.7 58.6 60.0 61.1 58.1 49.6 41.6 41.3
1937 40.8 41.0 38.4 47.4 54.1 58.6 61.4 61.8 56.3 50.9 41.4 37.1
1938 42.1 41.2 47.3 46.6 52.4 59.0 59.6 60.4 57.0 50.7 47.8 39.2
1939 39.4 40.9 42.4 47.8 52.4 58.0 60.7 61.8 58.2 46.7 46.6 37.8
```

Now we do a plot.

```
> plot(nottem)
```

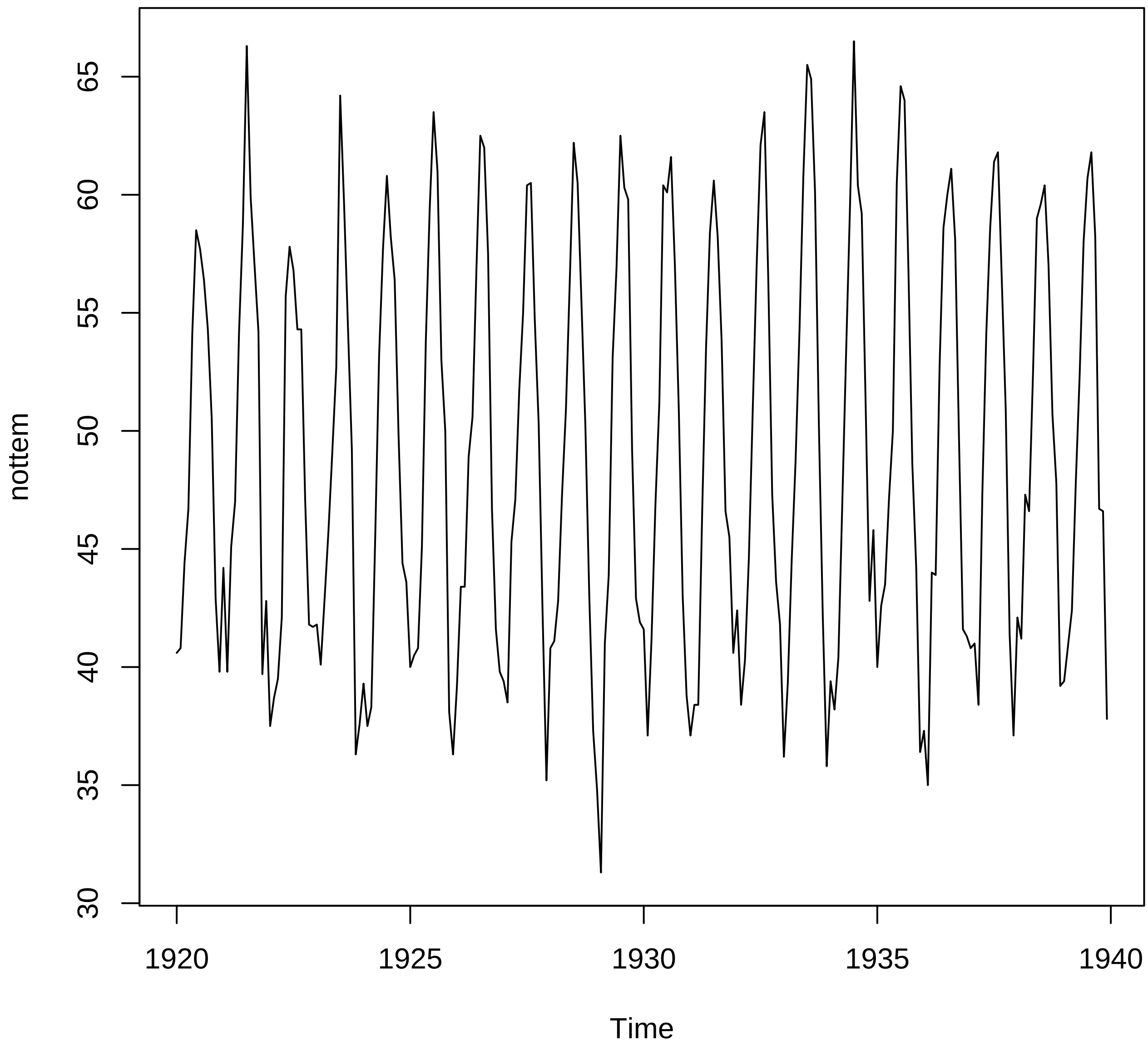
The plot is shown in Figure 2 where the temperature is in Fahrenheit.

Using the naked eye, it looks like the mean temperature is about 50 F. In fact, using **R** , it is

```
> mean(nottem)
[1] 49.03958
```

There are no upward nor downward trends but obviously there is a seasonality effect.

Figure 2: Plot of Monthly Nottingham Temperatures, 1920 – 1939



We can also get a summary of the temperatures from **R** .

```
> summary(nottem)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  31.30   41.55   47.35   49.04   57.00   66.50
```

Note that we can even use the help() command to find out more about the source of the data. Typing

```
help(nottem)
```

will bring up the references to the data.

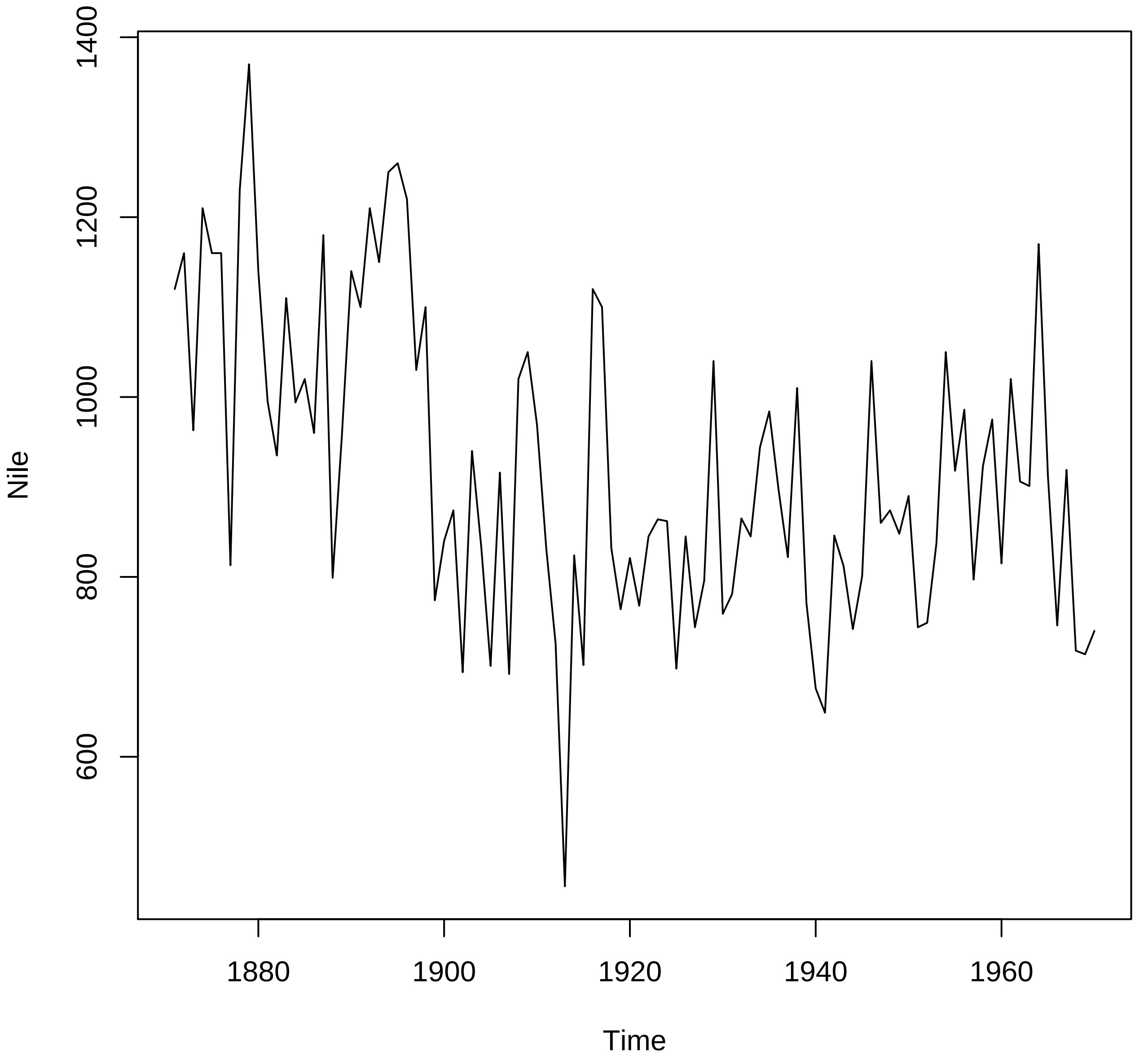
4. **Example 3**. Measurements of the annual flow of the river Nile at Aswan, 1871 – 1970, in $10^8 m^3$.

```
> Nile
Time Series:
Start = 1871
End = 1970
Frequency = 1
  [1] 1120 1160  963 1210 1160 1160  813 1230 1370 1140  995  935 1110  994 1020
 [16]  960 1180  799  958 1140 1100 1210 1150 1250 1260 1220 1030 1100  774  840
 [31]  874  694  940  833  701  916  692 1020 1050  969  831  726  456  824  702
 [46] 1120 1100  832  764  821  768  845  864  862  698  845  744  796 1040  759
 [61]  781  865  845  944  984  897  822 1010  771  676  649  846  812  742  801
 [76] 1040  860  874  848  890  744  749  838 1050  918  986  797  923  975  815
 [91] 1020  906  901 1170  912  746  919  718  714  740
```

We do a plot of the volume of the river flow.

From what we can see from the plot and also looking at the raw data, there appears to be a change in the annual flow around 1898 – 1899. Looking at the **R** display of the raw data, it seems that after the 28th observation (corresponding to year 1899), it is rarer to get annual flows above $1000 \times 10^8 m^3$. Perhaps that could be due to the building of a dam somewhere upstream.

Figure 3: Plot of Annual Flow of River Nile, 1871 – 1970 in $10^8 m^3$



There appears to be some sort of cyclical effect but the cycles are not regularly spaced compared to the data in Example 2 where we can expect the temperatures to rise and fall cyclically (seasonally) every 12 months.

In modelling this type of time series data, we will see later that we need to consider where the change point is.

4 Stochastic Processes

1. A time series is an example of a discrete time stochastic process.

2. A *discrete time stochastic process* is a sequence of random variables

   $$(X_1, X_2, \cdots).$$

   Sometimes the sequence terminates after a certain time, for example,

   $$(X_1, X_2, \cdots, X_n).$$

3. In general the random variables need not be identically distributed nor independent.

4. The X_i can also be continuous or discrete random variables. If they are continuous, we say that the process has continuous state space, otherwise it has discrete state space.

5. In fact, for most time series data, there is a degree of dependence between successive random variables, and the various time series models have been developed to account for the dependence.

6. In a lot of examples, individual terms $X_1, X_2, \cdots$ have the same mean and variance, in which case we say that the time series is stationary. So obviously if there is an upward trend or downward trend, the time series is not stationary. Likewise if there are seasonality effects, the time series is also not stationary.

7. However, in the course, we will learn how to remove the trend and/or seasonality, so that what remains is a stationary time series where we can estimate the parameters of the model.

8. **Example 4**. Louis Bachelier's stock pricing model.

   Louis Bachelier in his 1900 thesis *Théorie de la Spéculation* hypothesized that daily increments of stock prices are independently and identically distributed.

Let X_i be the increment of the stock price at the end of the ith day from the previous day. The value of X_i is negative if the stock price drops. Bachelier's model assume that all the X_i are independently and identically distributed as some normal random variable $N(\mu, \sigma)$. Thus we have a time series of daily stock increments

$$X_1, X_2, \cdots, X_n,$$

if we happen to track it to the nth day.

Since all the X_i have the same distribution, such a time series is stationary.

If the original starting stock price is S_0, then the stock price at the end of day k is

$$S_k = S_0 + \sum_{i=1}^{k} X_i = S_{k-1} + X_k.$$

It can be shown that the distribution of S_k is normal $N(S_0 + k\mu, \sqrt{k}\sigma)$.

Tutorial 1. Explore some of the datasets in **R** and pick up which ones are time series. If the data set is a time series, do a plot and comment on the data, for example, is it seasonal, is there a trend?